

## Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

**Product Description:** Di-tert-butyl dicarbonate, 70% w/w solution in THF  
**Cat No. :** S36750

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### 1.3. Details of the supplier of the safety data sheet

#### Company

Thermo Fisher (Kandel) GmbH  
Erlenbachweg 2, 76870 Kandel, Germany  
Tel: +49 (0) 721 84007 280  
Fax: +49 (0) 721 84007 300

**Swiss distributor** - Fisher Scientific AG  
Neuhofstrasse 11, CH 4153 Reinach  
Tel: +41 (0) 56 618 41 11  
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

**E-mail address** [begel.sdsdesk@thermofisher.com](mailto:begel.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:  
Tox Info Suisse Emergency Number: **145 (24hr)**  
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)  
Chemtrec (24h) Toll-Free: 0800 564 402  
Chemtrec Local: +41-43 508 20 11 (Zurich)

#### Poison Centre - Emergency information services

**Ireland** : National Poisons Information Centre (NPIC) -  
**01 809 2166** (8am-10pm, 7 days a week)  
**Malta** : +356 2395 2000  
**Cyprus** : +357 2240 5611

## Section 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

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## Physical hazards

Flammable liquids

Category 3 (H226)

## Health hazards

Acute oral toxicity

Category 4 (H302)

Acute Inhalation Toxicity - Vapors

Category 1 (H330)

Skin Corrosion/Irritation

Category 2 (H315)

Serious Eye Damage/Eye Irritation

Category 1 (H318)

Skin Sensitization

Category 1 (H317)

Carcinogenicity

Category 2 (H351)

Specific target organ toxicity - (single exposure)

Category 3 (H335) (H336)

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

## Hazard Statements

H226 - Flammable liquid and vapor

H302 - Harmful if swallowed

H330 - Fatal if inhaled

H315 - Causes skin irritation

H317 - May cause an allergic skin reaction

H318 - Causes serious eye damage

H335 - May cause respiratory irritation

H336 - May cause drowsiness or dizziness

H351 - Suspected of causing cancer

EUH019 - May form explosive peroxides

## Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P310 - Immediately call a POISON CENTER or doctor/physician

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P280 - Wear protective gloves/protective clothing/eye protection/face protection

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## 2.3. Other hazards

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## Section 3: Composition/information on ingredients

### 3.2. Mixtures

| Component                                       | CAS No     | EC No             | Weight % | CLP Classification - Regulation (EC) No 1272/2008                                                                                       |
|-------------------------------------------------|------------|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Dicarboxylic acid, bis(1,1-dimethylethyl) ester | 24424-99-5 | EEC No. 246-240-1 | 70       | Flam. Sol. 2 (H228)<br>STOT SE 3 (H335)<br>Skin Irrit. 2 (H315)<br>Eye Dam. 1 (H318)<br>Skin Sens. 1 (H317)<br>Acute Tox. 1 (H330)      |
| Tetrahydrofuran                                 | 109-99-9   | 203-726-8         | 30       | Flam. Liq. 2 (H225)<br>Acute Tox. 4 (H302)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)<br>STOT SE 3 (H336)<br>Carc. 2 (H351)<br>(EUH019) |

| Component       | Specific concentration limits (SCL's)                                    | M-Factor | Component notes |
|-----------------|--------------------------------------------------------------------------|----------|-----------------|
| Tetrahydrofuran | Acute Tox. 4 :: C>82.5%<br>Eye Irrit. 2 :: C>=25%<br>STOT SE 3 :: C>=25% | -        | -               |

Full text of Hazard Statements: see section 16

## Section 4: First aid measures

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                     |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                    |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                 |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                               |

### 4.2. Most important symptoms and effects, both acute and delayed

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None reasonably foreseeable. . Causes severe eye damage. May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## 4.3. Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## Section 5: Firefighting measures

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

No information available.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### 6.2. Environmental precautions

Should not be released into the environment.

### 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7: Handling and storage

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## 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. If peroxide formation is suspected, do not open or move container. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals. Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Technical Rules for Hazardous Substances (TRGS) 510**  
**Storage Class (LGK) (Germany)**

Class 3

**Switzerland - Storage of hazardous substances**

Storage class - SC 3  
<https://www.kvu.ch/de/themen/stoffe-und-produkte>  
<https://www.kvu.ch/fr/themes/substances-et-produits>  
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

## 7.3. Specific end use(s)

Use in laboratories

## Section 8: Exposure controls/personal protection

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund).

| Component       | European Union                                                                                                              | The United Kingdom                                                                                                        | France                                                                                                                                                                                                                         | Belgium                                                                                                                               | Spain                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm (8h)<br>TWA: 150 mg/m <sup>3</sup> (8h)<br>STEL: 100 ppm (15min)<br>STEL: 300 mg/m <sup>3</sup> (15min)<br>Skin | STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15 min<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>Skin | TWA / VME: 50 ppm (8 heures). restrictive limit<br>TWA / VME: 150 mg/m <sup>3</sup> (8 heures). restrictive limit<br>STEL / VLCT: 100 ppm. restrictive limit<br>STEL / VLCT: 300 mg/m <sup>3</sup> . restrictive limit<br>Peau | TWA: 50 ppm 8 uren<br>TWA: 150 mg/m <sup>3</sup> 8 uren<br>STEL: 100 ppm 15 minuten<br>STEL: 300 mg/m <sup>3</sup> 15 minuten<br>Huid | STEL / VLA-EC: 100 ppm (15 minutos).<br>STEL / VLA-EC: 300 mg/m <sup>3</sup> (15 minutos).<br>TWA / VLA-ED: 50 ppm (8 horas)<br>TWA / VLA-ED: 150 mg/m <sup>3</sup> (8 horas)<br>Piel |

| Component       | Italy                                                                                                                         | Germany                                                                                           | Portugal                                                                                  | The Netherlands                                                            | Finland                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50 ppm 8 ore.<br>Time Weighted Average<br>TWA: 150 mg/m <sup>3</sup> 8 ore.<br>Time Weighted Average<br>STEL: 100 ppm 15 | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 150 mg/m <sup>3</sup> (8 Stunden). AGW - | STEL: 100 ppm 15 minutos<br>STEL: 300 mg/m <sup>3</sup> 15 minutos<br>TWA: 50 ppm 8 horas | huid<br>STEL: 200 ppm 15 minuten<br>STEL: 600 mg/m <sup>3</sup> 15 minuten | TWA: 50 ppm 8 tunteina<br>TWA: 150 mg/m <sup>3</sup> 8 tunteina<br>STEL: 100 ppm 15 minuutteina |

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|  |                                                                                     |                                                                                                                                                                        |                                               |                                                          |                                                      |
|--|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------|------------------------------------------------------|
|  | minuti. Short-term<br>STEL: 300 mg/m <sup>3</sup> 15<br>minuti. Short-term<br>Pelle | exposure factor 2<br>TWA: 20 ppm (8<br>Stunden). MAK<br>TWA: 60 mg/m <sup>3</sup> (8<br>Stunden). MAK<br>Höhepunkt: 40 ppm<br>Höhepunkt: 120 mg/m <sup>3</sup><br>Haut | TWA: 150 mg/m <sup>3</sup> 8<br>horas<br>Pele | TWA: 100 ppm 8 uren<br>TWA: 300 mg/m <sup>3</sup> 8 uren | STEL: 300 mg/m <sup>3</sup> 15<br>minuutteina<br>Iho |
|--|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------|------------------------------------------------------|

| Component       | Austria                                                                                                                                                                 | Denmark                                                                                                                                        | Switzerland                                                                                                                                                  | Poland                                                                                  | Norway                                                                                                                                                                                    |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | Haut<br>MAK-KZGW: 100 ppm<br>15 Minuten<br>MAK-KZGW: 300 mg/m <sup>3</sup><br>15 Minuten<br>MAK-TMW: 50 ppm 8<br>Stunden<br>MAK-TMW: 150 mg/m <sup>3</sup><br>8 Stunden | TWA: 50 ppm 8 timer<br>TWA: 150 mg/m <sup>3</sup> 8 timer<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutter<br>STEL: 100 ppm 15<br>minutter<br>Hud | Haut/Peau<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 300 mg/m <sup>3</sup> 15<br>Minuten<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 150 mg/m <sup>3</sup> 8<br>Stunden | STEL: 300 mg/m <sup>3</sup> 15<br>minutach<br>TWA: 150 mg/m <sup>3</sup> 8<br>godzinach | TWA: 50 ppm 8 timer<br>TWA: 150 mg/m <sup>3</sup> 8 timer<br>STEL: 75 ppm 15<br>minutter. value<br>calculated<br>STEL: 187.5 mg/m <sup>3</sup> 15<br>minutter. value<br>calculated<br>Hud |

| Component       | Bulgaria                                                                                                           | Croatia                                                                                                                                                                       | Ireland                                                                                                                        | Cyprus                                                                                                                                  | Czech Republic                                                                                                       |
|-----------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | TWA: 50.0 ppm<br>TWA: 150.0 mg/m <sup>3</sup><br>STEL : 100 ppm<br>STEL : 300.0 mg/m <sup>3</sup><br>Skin notation | kože<br>TWA-GVI: 50 ppm 8<br>satima.<br>TWA-GVI: 150 mg/m <sup>3</sup> 8<br>satima.<br>STEL-KGVI: 100 ppm<br>15 minutama.<br>STEL-KGVI: 300 mg/m <sup>3</sup><br>15 minutama. | TWA: 50 ppm 8 hr.<br>TWA: 150 mg/m <sup>3</sup> 8 hr.<br>STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15<br>min<br>Skin | Skin-potential for<br>cutaneous absorption<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 150 mg/m <sup>3</sup> 8<br>hodinách.<br>Potential for cutaneous<br>absorption<br>Ceiling: 300 mg/m <sup>3</sup> |

| Component       | Estonia                                                                                                                                                           | Gibraltar                                                                                                                             | Greece                                                                                     | Hungary                                                                                                                                                                                                         | Iceland                                                                                                                                            |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | Nahk<br>TWA: 50 ppm 8<br>tundides.<br>TWA: 150 mg/m <sup>3</sup> 8<br>tundides.<br>STEL: 100 ppm 15<br>minutites.<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutites. | Skin notation<br>TWA: 50 ppm 8 hr<br>TWA: 150 mg/m <sup>3</sup> 8 hr<br>STEL: 100 ppm 15 min<br>STEL: 300 mg/m <sup>3</sup> 15<br>min | STEL: 250 ppm<br>STEL: 735 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 590 mg/m <sup>3</sup> | STEL: 300 mg/m <sup>3</sup> 15<br>percekben. CK<br>STEL: 100 ppm 15<br>percekben. CK<br>TWA: 150 mg/m <sup>3</sup> 8<br>óraban. AK<br>TWA: 50 ppm 8 óraban.<br>AK<br>lehetséges borón<br>keresztüli felszívódás | STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm 8<br>klukkustundum.<br>TWA: 150 mg/m <sup>3</sup> 8<br>klukkustundum.<br>Skin notation |

| Component       | Latvia                                                                                                                                  | Lithuania                                                                                                  | Luxembourg                                                                                                                                                                                               | Malta                                                                                                                                                                        | Romania                                                                                                                                          |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | skin - potential for<br>cutaneous exposure<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup><br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 50 ppm IPRD<br>TWA: 150 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 100 ppm<br>STEL: 300 mg/m <sup>3</sup> | Possibility of significant<br>uptake through the skin<br>TWA: 50 ppm 8<br>Stunden<br>TWA: 150 mg/m <sup>3</sup> 8<br>Stunden<br>STEL: 100 ppm 15<br>Minuten<br>STEL: 300 mg/m <sup>3</sup> 15<br>Minuten | possibility of significant<br>uptake through the skin<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup><br>STEL: 100 ppm 15<br>minuti<br>STEL: 300 mg/m <sup>3</sup> 15<br>minuti | Skin notation<br>TWA: 50 ppm 8 ore<br>TWA: 150 mg/m <sup>3</sup> 8 ore<br>STEL: 100 ppm 15<br>minute<br>STEL: 300 mg/m <sup>3</sup> 15<br>minute |

| Component       | Russia                     | Slovak Republic                                                                                                      | Slovenia                                                                                                                                    | Sweden                                                                                                                                                                  | Turkey                                                                                                                                    |
|-----------------|----------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Tetrahydrofuran | MAC: 100 mg/m <sup>3</sup> | Ceiling: 300 mg/m <sup>3</sup><br>Potential for cutaneous<br>absorption<br>TWA: 50 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 50 ppm 8 urah<br>TWA: 150 mg/m <sup>3</sup> 8 urah<br>Koža<br>STEL: 100 ppm 15<br>minutah<br>STEL: 300 mg/m <sup>3</sup> 15<br>minutah | Binding STEL: 100 ppm<br>15 minuter<br>Binding STEL: 300<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 50 ppm 8 timmar.<br>NGV<br>TLV: 150 mg/m <sup>3</sup> 8<br>timmar. NGV | Deri<br>TWA: 50 ppm 8 saat<br>TWA: 150 mg/m <sup>3</sup> 8 saat<br>STEL: 100 ppm 15<br>dakika<br>STEL: 300 mg/m <sup>3</sup> 15<br>dakika |

## Biological limit values

List source(s):

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| Component       | European Union | United Kingdom | France | Spain                                         | Germany                                          |
|-----------------|----------------|----------------|--------|-----------------------------------------------|--------------------------------------------------|
| Tetrahydrofuran |                |                |        | Tetrahydrofuran: 2 mg/L<br>urine end of shift | Tetrahydrofuran: 2 mg/L<br>urine (end of shift ) |

| Component       | Gibraltar | Latvia | Slovak Republic                                                   | Luxembourg | Turkey |
|-----------------|-----------|--------|-------------------------------------------------------------------|------------|--------|
| Tetrahydrofuran |           |        | Tetrahydrofuran: 2 mg/L<br>urine end of exposure or<br>work shift |            |        |

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

MDHS 88 Volatile organic compounds in air. Laboratory method using diffusive samplers, solvent desorption and gas chromatography

MDHS 96 Volatile organic compounds in air - Laboratory method using pumped solid sorbent tubes, solvent desorption and gas chromatography

## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                                             | Acute effects local<br>(Dermal) | Acute effects<br>systemic (Dermal) | Chronic effects local<br>(Dermal) | Chronic effects<br>systemic (Dermal) |
|-----------------------------------------------------------------------|---------------------------------|------------------------------------|-----------------------------------|--------------------------------------|
| Dicarbonic acid,<br>bis(1,1-dimethylethyl) ester<br>24424-99-5 ( 70 ) |                                 |                                    |                                   | DNEL = 686µg/kg<br>bw/day            |
| Tetrahydrofuran<br>109-99-9 ( 30 )                                    |                                 |                                    |                                   | DNEL = 12.6mg/kg<br>bw/day           |

| Component                                                             | Acute effects local<br>(Inhalation) | Acute effects<br>systemic (Inhalation) | Chronic effects local<br>(Inhalation) | Chronic effects<br>systemic (Inhalation) |
|-----------------------------------------------------------------------|-------------------------------------|----------------------------------------|---------------------------------------|------------------------------------------|
| Dicarbonic acid,<br>bis(1,1-dimethylethyl) ester<br>24424-99-5 ( 70 ) |                                     |                                        |                                       | DNEL = 2.42mg/m <sup>3</sup>             |
| Tetrahydrofuran<br>109-99-9 ( 30 )                                    | DNEL = 300mg/m <sup>3</sup>         | DNEL = 96mg/m <sup>3</sup>             | DNEL = 150mg/m <sup>3</sup>           | DNEL = 72.4mg/m <sup>3</sup>             |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                                                             | Fresh water     | Fresh water<br>sediment         | Water Intermittent | Microorganisms in<br>sewage treatment | Soil (Agriculture)          |
|-----------------------------------------------------------------------|-----------------|---------------------------------|--------------------|---------------------------------------|-----------------------------|
| Dicarbonic acid,<br>bis(1,1-dimethylethyl) ester<br>24424-99-5 ( 70 ) | PNEC = 1.25mg/L | PNEC = 10.8mg/kg<br>sediment dw | PNEC = 12.5mg/L    | PNEC = 30.7mg/L                       | PNEC = 3.92mg/kg<br>soil dw |
| Tetrahydrofuran<br>109-99-9 ( 30 )                                    | PNEC = 4.32mg/L | PNEC = 23.3mg/kg<br>sediment dw | PNEC = 21.6mg/L    | PNEC = 4.6mg/L                        | PNEC = 2.13mg/kg<br>soil dw |

| Component                                                             | Marine water     | Marine water<br>sediment        | Marine water<br>Intermittent | Food chain             | Air |
|-----------------------------------------------------------------------|------------------|---------------------------------|------------------------------|------------------------|-----|
| Dicarbonic acid,<br>bis(1,1-dimethylethyl) ester<br>24424-99-5 ( 70 ) | PNEC = 0.125mg/L | PNEC = 1.08mg/kg<br>sediment dw |                              |                        |     |
| Tetrahydrofuran<br>109-99-9 ( 30 )                                    | PNEC = 0.432mg/L | PNEC = 2.33mg/kg<br>sediment dw |                              | PNEC = 67mg/kg<br>food |     |

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## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material  | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Nitrile rubber  | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Neoprene        |                                   |                 |             |                       |
| Natural rubber  |                                   |                 |             |                       |
| PVC             |                                   |                 |             |                       |
| Butyl rubber    |                                   |                 |             |                       |
| Neoprene gloves |                                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

#### Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

No information available.

## Section 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

#### Physical State

Liquid

#### Appearance

##### Odor

No information available

##### Odor Threshold

No data available

##### Melting Point/Range

No data available

##### Softening Point

No data available

##### Boiling Point/Range

No information available

##### Flammability (liquid)

Flammable

Estimated

##### Flammability (solid,gas)

Not applicable

Liquid



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|                                                 |                          |                                   |
|-------------------------------------------------|--------------------------|-----------------------------------|
| Explosion Limits                                | No data available        |                                   |
| Flash Point                                     | No information available | Method - No information available |
| Autoignition Temperature                        | No data available        |                                   |
| Decomposition Temperature                       | No data available        |                                   |
| pH                                              | Not applicable           |                                   |
| Viscosity                                       | No data available        |                                   |
| Water Solubility                                | No information available |                                   |
| Solubility in other solvents                    | No information available |                                   |
| Partition Coefficient (n-octanol/water)         |                          |                                   |
| Component                                       | log Pow                  |                                   |
| Dicarboxylic acid, bis(1,1-dimethylethyl) ester | 1.87                     |                                   |
| Tetrahydrofuran                                 | 0.45                     |                                   |
| Vapor Pressure                                  | No data available        |                                   |
| Density / Specific Gravity                      | No data available        |                                   |
| Bulk Density                                    | Not applicable           | Liquid                            |
| Vapor Density                                   | No data available        | (Air = 1.0)                       |
| Particle characteristics                        | (liquid) Not applicable  |                                   |

## 9.2. Other information

Explosive Properties explosive air/vapour mixtures possible

## Section 10: Stability and reactivity

10.1. Reactivity None known, based on information available

10.2. Chemical stability Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization No information available.  
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid Keep away from open flames, hot surfaces and sources of ignition.

10.5. Incompatible materials . Strong oxidizing agents.

10.6. Hazardous decomposition products Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## Section 11: Toxicological information

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

(a) acute toxicity;  
Oral Category 4  
Dermal Based on available data, the classification criteria are not met  
Inhalation Category 1

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## Toxicology data for the components

| Component                                       | LD50 Oral              | LD50 Dermal            | LC50 Inhalation                               |
|-------------------------------------------------|------------------------|------------------------|-----------------------------------------------|
| Dicarboxylic acid, bis(1,1-dimethylethyl) ester | >5000 mg/kg ( Mammal ) | >2000 mg/kg ( Mammal ) | LC50 = 100 mg/m <sup>3</sup> ( Rat ) 4 h      |
| Tetrahydrofuran                                 | 1650 mg/kg ( Rat )     | > 2000 mg/kg (Rabbit)  | 180 mg/L ( Rat ) 1 h<br>53.9 mg/L ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin Category 1

| Component                          | Test method                                       | Test species | Study result    |
|------------------------------------|---------------------------------------------------|--------------|-----------------|
| Tetrahydrofuran<br>109-99-9 ( 30 ) | Local Lymph Node Assay<br>OECD Test Guideline 429 | mouse        | non-sensitising |

May cause sensitization by skin contact

(e) germ cell mutagenicity; No data available

| Component                          | Test method                                             | Test species          | Study result |
|------------------------------------|---------------------------------------------------------|-----------------------|--------------|
| Tetrahydrofuran<br>109-99-9 ( 30 ) | OECD Test Guideline 476<br>Gene cell mutation           | in vivo<br>Mammalian  | negative     |
|                                    | OECD Test Guideline 473<br>Chromosomal aberration assay | in vitro<br>Mammalian | negative     |

(f) carcinogenicity; Category 2

The table below indicates whether each agency has listed any ingredient as a carcinogen  
Limited evidence of a carcinogenic effect

| Component       | EU | UK | Germany | IARC     |
|-----------------|----|----|---------|----------|
| Tetrahydrofuran |    |    |         | Group 2B |

(g) reproductive toxicity; No data available

| Component                          | Test method             | Test species / Duration | Study result      |
|------------------------------------|-------------------------|-------------------------|-------------------|
| Tetrahydrofuran<br>109-99-9 ( 30 ) | OECD Test Guideline 416 | Rat<br>2 Generation     | NOAEL = 3,000 ppm |

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system, Central nervous system (CNS).

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.  
Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

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## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## Section 12: Ecological information

### 12.1. Toxicity

#### Ecotoxicity effects

| Component       | Freshwater Fish                                                                     | Water Flea                                   | Freshwater Algae |
|-----------------|-------------------------------------------------------------------------------------|----------------------------------------------|------------------|
| Tetrahydrofuran | 2160 mg/l LC50 = 96 h<br>Pimephales promelas<br>Leuciscus idus: LC50: 2820 mg/L/48h | EC50 48 h 3485 mg/l<br>EC50: >10000 mg/L/24h |                  |

**12.2. Persistence and degradability** No information available

**12.3. Bioaccumulative potential** No information available

| Component                                     | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------------------|---------|-------------------------------|
| Dicarboxic acid, bis(1,1-dimethylethyl) ester | 1.87    | No data available             |
| Tetrahydrofuran                               | 0.45    | No data available             |

**12.4. Mobility in soil** No information available

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

### 12.6. Endocrine disrupting properties

#### Endocrine Disruptor Information

| Component       | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances |
|-----------------|------------------------------------------|--------------------------------------------------|
| Tetrahydrofuran | Group III Chemical                       |                                                  |

### 12.7. Other adverse effects

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13: Disposal considerations

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

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|                                       |                                                                                                                                                                                                                                                                                                            |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>European Waste Catalogue (EWC)</b> | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                                                                                                   |
| <b>Other Information</b>              | Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains.                                                                          |
| <b>Switzerland - Waste Ordinance</b>  | Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600<br><a href="https://www.fedlex.admin.ch/eli/cc/2015/891/en">https://www.fedlex.admin.ch/eli/cc/2015/891/en</a> |

## Section 14: Transport information

### IMDG/IMO

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
| <b>14.1. UN number</b>                  | UN2929                                   |
| <b>14.2. UN proper shipping name</b>    | TOXIC LIQUID, FLAMMABLE, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b>          | Di-tert-butyl dicarbonate/THF            |
| <b>14.3. Transport hazard class(es)</b> | 6.1                                      |
| <b>Subsidiary Hazard Class</b>          | 3                                        |
| <b>14.4. Packing group</b>              | II                                       |

### ADR

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
| <b>14.1. UN number</b>                  | UN2929                                   |
| <b>14.2. UN proper shipping name</b>    | TOXIC LIQUID, FLAMMABLE, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b>          | Di-tert-butyl dicarbonate/THF            |
| <b>14.3. Transport hazard class(es)</b> | 6.1                                      |
| <b>Subsidiary Hazard Class</b>          | 3                                        |
| <b>14.4. Packing group</b>              | II                                       |

### IATA

|                                         |                                          |
|-----------------------------------------|------------------------------------------|
| <b>14.1. UN number</b>                  | UN2929                                   |
| <b>14.2. UN proper shipping name</b>    | TOXIC LIQUID, FLAMMABLE, ORGANIC, N.O.S. |
| <b>Technical Shipping Name</b>          | Di-tert-butyl dicarbonate/THF            |
| <b>14.3. Transport hazard class(es)</b> | 6.1                                      |
| <b>Subsidiary Hazard Class</b>          | 3                                        |
| <b>14.4. Packing group</b>              | II                                       |

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| <b>14.5. Environmental hazards</b>                                   | No hazards identified            |
| <b>14.6. Special precautions for user</b>                            | No special precautions required. |
| <b>14.7. Maritime transport in bulk according to IMO instruments</b> | Not applicable, packaged goods   |

## Section 15: Regulatory information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No | EINECS | ELINCS | NLP | IECSC | TCSI | KECL | ENCS | ISHL |
|-----------|--------|--------|--------|-----|-------|------|------|------|------|
|-----------|--------|--------|--------|-----|-------|------|------|------|------|

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|                                               |            |           |   |   |   |   |          |   |   |
|-----------------------------------------------|------------|-----------|---|---|---|---|----------|---|---|
| Dicarbonic acid, bis(1,1-dimethylethyl) ester | 24424-99-5 | 246-240-1 | - | - | X | X | KE-10019 | X | X |
| Tetrahydrofuran                               | 109-99-9   | 203-726-8 | - | - | X | X | KE-33454 | X | X |

| Component                                     | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------------------------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Dicarbonic acid, bis(1,1-dimethylethyl) ester | 24424-99-5 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Tetrahydrofuran                               | 109-99-9   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                                     | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Dicarbonic acid, bis(1,1-dimethylethyl) ester | 24424-99-5 | -                                                                   | -                                                                             | -                                                                                                     |
| Tetrahydrofuran                               | 109-99-9   | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                                     | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Dicarbonic acid, bis(1,1-dimethylethyl) ester | 24424-99-5 | Not applicable                                                                            | Not applicable                                                                           |
| Tetrahydrofuran                               | 109-99-9   | Not applicable                                                                            | Not applicable                                                                           |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = 1 (self classification)

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------|---------------------------------------|-------------------------|
| Tetrahydrofuran | WGK1                                  |                         |

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| Component       | France - INRS (Tables of occupational diseases)      |
|-----------------|------------------------------------------------------|
| Tetrahydrofuran | Tableaux des maladies professionnelles (TMP) - RG 84 |

## Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

| Component                          | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Tetrahydrofuran<br>109-99-9 ( 30 ) |                                                                                                                | Group I                                                                         |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## Section 16: Other information

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed  
H330 - Fatal if inhaled  
H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H318 - Causes serious eye damage  
H335 - May cause respiratory irritation  
H336 - May cause drowsiness or dizziness  
H351 - Suspected of causing cancer  
EUH019 - May form explosive peroxides  
H225 - Highly flammable liquid and vapor  
H228 - Flammable solid  
H319 - Causes serious eye irritation

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (volatile organic compound)

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Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

**Prepared By** Health, Safety and Environmental Department

**Revision Date** 07-Dec-2024

**Revision Summary** Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,  
Number 3, Chemo (SR 813.11 - Ordinance on Protection against Dangerous Substances and  
Preparations).**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**